Sampling Distributions & Central Limit Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: What is a sampling distribution?**

Explain the difference between a population distribution and the sampling distribution of a sample mean $\bar{x}$.

The population distribution describes the values of individual members of the entire population. The sampling distribution of $\bar{x}$ describes the values that the sample mean would take if you repeatedly took many samples of the same size from that population and computed the mean of each sample — it describes variability between different samples' means, not between individuals.

2 **[Foundational] • Target: What is a sampling distribution?**

Explain why the sampling distribution of $\bar{x}$ has less variability than the original population distribution.

A sample mean averages together multiple individual values, and averaging tends to smooth out extreme individual values — some will be higher and some lower than the true mean, partially canceling out. This averaging effect makes sample means cluster more tightly around the population mean than individual values do.

3 **[Foundational] • Target: Mean and standard deviation of the sampling distribution**

A population has mean $\mu = 60$ and standard deviation $\sigma = 12$. For samples of size $n = 36$, find the mean and standard deviation of the sampling distribution of $\bar{x}$.

Mean of $\bar{x}$: $\mu_{\bar{x}} = \mu = 60$ (unbiased — the sampling distribution's mean equals the population mean). Standard deviation of $\bar{x}$ (standard error): $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$.

4 **[Foundational] • Target: Mean and standard deviation of the sampling distribution**

Using the same population, find the standard error for a larger sample of size $n = 144$, and explain what happens to the standard error as sample size increases.

$\sigma_{\bar{x}} = \frac{12}{\sqrt{144}} = \frac{12}{12} = 1$. As sample size increases, the standard error decreases (specifically, it decreases proportionally to $\frac{1}{\sqrt{n}}$) — larger samples give more precise (less variable) estimates of the population mean.

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5 [Foundational] • Target: The Central Limit Theorem

State the Central Limit Theorem in your own words, including what happens to the shape of the sampling distribution of $\bar{x}$ as sample size n increases.

The Central Limit Theorem states that as sample size n increases, the sampling distribution of the sample mean $\bar{x}$ becomes approximately normal, regardless of the shape of the original population distribution — even if the population itself is skewed or unusual in shape. A commonly used guideline is that $n \geq 30$ is generally large enough for this approximation to be reasonable.

6 [Foundational] • Target: The Central Limit Theorem

Explain why the Central Limit Theorem is considered one of the most important results in statistics, referencing its practical usefulness.

The Central Limit Theorem is powerful because it allows statisticians to use normal-distribution-based methods (like z-scores and confidence intervals) for the sample mean even when the underlying population distribution is unknown or clearly not normal — as long as the sample size is reasonably large, this makes a huge range of inference procedures broadly applicable.

7 [Foundational] • Target: Applying the Central Limit Theorem

A population of package weights has mean $\mu = 50$ lbs and standard deviation $\sigma = 8$ lbs (population shape unknown). For a random sample of $n = 64$ packages, find the probability the sample mean weight exceeds 52 lbs, given $P(Z > 2) \approx 0.0228$.

Since $n = 64 \geq 30$, the CLT justifies treating $\bar{x}$ as approximately normal. $\sigma_{\bar{x}} = \frac{8}{\sqrt{64}} = \frac{8}{8} = 1$. $z = \frac{52 - 50}{1} = 2$. $P(\bar{x} > 52) = P(Z > 2) \approx 0.0228$, or about 2.28%.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sampling distribution of a sample proportion

A population has a true proportion $p = 0.4$ favoring a proposal. For samples of size $n = 100$, find the mean and standard deviation of the sampling distribution of the sample proportion $\hat{p}$, using $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.

Mean: $\mu_{\hat{p}} = p = 0.4$. Standard deviation: $\sigma_{\hat{p}} = \sqrt{\frac{0.4(0.6)}{100}} = \sqrt{\frac{0.24}{100}} = \sqrt{0.0024} \approx 0.049$.

9 [Intermediate] • Target: Sampling distribution of a sample proportion

State the two conditions typically required to treat the sampling distribution of $\hat{p}$ as approximately normal.

Both $np \geq 10$ and $n(1 - p) \geq 10$ must hold (sometimes stated as needing at least 10 expected successes and 10 expected failures) — this ensures the sampling distribution of $\hat{p}$ is not too skewed to approximate with a normal curve.

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10 **[Intermediate]** • **Target: Comparing sampling variability across sample sizes**

Explain, in the context of opinion polling, why a poll of 1000 people gives a more reliable estimate of a population proportion than a poll of 50 people, connecting your answer to the formula for $\sigma_{\hat{p}}$.

Since $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ decreases as n increases, a larger sample size produces a smaller standard error for the sample proportion — meaning the sample proportion from a poll of 1000 people tends to land closer to the true population proportion, on average, than a poll of only 50 people, which is subject to much more sample-to-sample variability.

11 **[Intermediate]** • **Target: Applying the sampling distribution of the sample mean**

A population of commute times has mean $\mu = 32$ minutes and standard deviation $\sigma = 10$ minutes. For a random sample of $n = 25$ commuters, find the probability the sample mean commute time exceeds 35 minutes, given $P(Z > 1.5) \approx 0.0668$.

$\sigma_{\bar{x}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$. $z = \frac{35-32}{2} = 1.5$. $P(\bar{x} > 35) = P(Z > 1.5) \approx 0.0668$, or about 6.68%.

12 **[Intermediate]** • **Target: Sample size and the CLT threshold**

A population of insurance claim amounts is strongly right-skewed. Explain why a sample of size $n = 15$ might not be large enough to safely use CLT-based normal methods, but a sample of size $n = 200$ likely would be.

The Central Limit Theorem's normal approximation improves as sample size increases, and it takes a larger sample size to overcome a strongly skewed population shape. With only $n = 15$, the sampling distribution of $\bar{x}$ may still noticeably reflect the skewness of the original population, making normal-based methods unreliable. With $n = 200$ (well beyond the common $n \geq 30$ guideline), the sampling distribution of $\bar{x}$ is much more likely to be close to normal, even though the original population itself is skewed.

13 **[Intermediate]** • **Target: Standard error vs. standard deviation**

Explain the difference between the standard deviation of an individual population value and the standard error of a sample mean $\bar{x}$.

The standard deviation (σ) measures the spread of individual values within the population. The standard error ($\sigma_{\bar{x}} = \sigma/\sqrt{n}$) measures the spread of sample means across many hypothetical samples of the same size — it is always smaller than σ (for $n > 1$), since averaging reduces variability.

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14 **[Intermediate]** • **Target: Standard error vs. standard deviation**

A quality control engineer at a bottling plant knows that individual bottle fill volumes are approximately normal with mean 500 mL and standard deviation 8 mL. Every hour, the engineer measures a sample of 16 bottles and computes the sample mean fill volume. Describe the shape, mean, and standard deviation of the sampling distribution of this hourly sample mean, and explain whether the Central Limit Theorem's usual $n \geq 30$ guideline is even needed to justify treating this sampling distribution as normal.

Shape: approximately normal. Mean: $\mu_{\bar{x}} = 500$ mL (same as the population mean). Standard deviation: $\sigma_{\bar{x}} = \frac{8}{\sqrt{16}} = \frac{8}{4} = 2$ mL. The $n \geq 30$ CLT guideline is not needed here, because the underlying population of individual fill volumes is already stated to be approximately normal — when the population itself is normal, the sampling distribution of $\bar{x}$ is exactly normal for any sample size, including small ones like $n = 16$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Standard error vs. standard deviation** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

A political pollster samples 900 voters and calculates a sample proportion $\hat{p} = 0.53$ supporting a ballot measure, from a true population proportion of $p = 0.50$. Verify that both conditions ($np \geq 10$ and $n(1-p) \geq 10$) are satisfied for treating the sampling distribution of $\hat{p}$ as approximately normal, then find the standard deviation of this sampling distribution, to four decimal places.

$np = 900(0.50) = 450 \geq 10$ ✓. $n(1-p) = 900(0.50) = 450 \geq 10$ ✓. Both conditions are satisfied. Standard deviation: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.50)(0.50)}{900}} = \sqrt{\frac{0.25}{900}} \approx 0.0167$.

16 **[Challenging]** • **Target: Standard error vs. standard deviation** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A large factory produces light bulbs whose individual lifespans are strongly right-skewed (most bulbs fail early, but a few last an unusually long time), with population mean $\mu = 800$ hours and standard deviation $\sigma = 250$ hours. An engineer takes a random sample of $n = 64$ bulbs and computes the sample mean lifespan. Explain, step by step, why it is reasonable to treat this sampling distribution of $\bar{x}$ as approximately normal despite the population itself being skewed, then compute the standard error of $\bar{x}$.

Even though the individual bulb lifespans are strongly skewed, the Central Limit Theorem states that the sampling distribution of the sample mean becomes approximately normal as sample size increases, regardless of the population's original shape. Since $n = 64$ comfortably exceeds the common guideline of $n \geq 30$, it is reasonable to treat $\bar{x}$'s sampling distribution as approximately normal here. Standard error: $\sigma_{\bar{x}} = \frac{250}{\sqrt{64}} = \frac{250}{8} = 31.25$ hours.

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17 **[Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

An online retailer knows that 22% of all website visitors make a purchase. On a particular day, the retailer records the browsing sessions of a random sample of 500 visitors and calculates the sample proportion who made a purchase. Verify the two conditions needed to treat this sampling distribution as approximately normal, then compute the standard deviation of the sampling distribution of the sample proportion, to four decimal places, and describe what this standard deviation represents in the context of comparing many different days' worth of visitor samples.

$np = 500(0.22) = 110 \geq 10$ ✓. $n(1-p) = 500(0.78) = 390 \geq 10$ ✓. Both conditions are satisfied. $\sigma_{\hat{p}} = \sqrt{\frac{(0.22)(0.78)}{500}} = \sqrt{\frac{0.1716}{500}} = \sqrt{0.0003432} \approx 0.0185$. This standard deviation describes how much the sample proportion of purchasers would typically vary from one day's random sample of 500 visitors to another, if the true purchase rate stayed fixed at 22% — most daily sample proportions would fall within roughly ± 0.0185 of 0.22.

18 **[Challenging] • Target: Sampling distribution of a sample proportion • Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

The distribution of cholesterol levels in a population is approximately normal with $\mu = 190$ mg/dL and $\sigma = 24$ mg/dL. (a) What proportion of individuals have cholesterol levels above 226 mg/dL? (b) What cholesterol level marks the 90th percentile?

(a) Proportion Above 226: Standardize: $z = \frac{226 - 190}{24} = \frac{36}{24} = 1.50$. Using standard normal tables: $P(Z > 1.50) = 1 - 0.9332 = \mathbf{0.0668}$ (6.68%). (b) 90th Percentile: Find z^* such that $P(Z < z^*) = 0.90$, which gives $z^* \approx 1.282$. Unstandardize: $x = \mu + z^*\sigma = 190 + 1.282(24) = 190 + 30.77 = \mathbf{220.77}$ mg/dL.

19 **[Challenging] • Target: Comparing sampling variability across sample sizes • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A histogram is used to display the distribution of the number of hours college students reported spending on social media per day. The bins for the histogram are as follows: 0-2 hours, 2-4 hours, 4-6 hours, 6-8 hours, and 8-10 hours. The heights of the bars are 5, 12, 15, 8, and 3, respectively. What is the total number of students represented by this histogram?

The heights of the bars in a histogram represent the frequencies (counts) of data points within each bin. To find the total number of students represented, we sum the heights of all the bars. Total students = Height of bar 1 + Height of bar 2 + Height of bar 3 + Height of bar 4 + Height of bar 5 Total students = 5 + 12 + 15 + 8 + 3 = 43. Therefore, 43 students are represented by this histogram.

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20 [Challenging] • Target: Applying the sampling distribution of the sample mean • Cross-Lesson: 3.03 The Normal Distribution

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

1. State: Test $H_0 : \mu = 200$ g vs $H_a : \mu < 200$ g at $\alpha = 0.05$, where μ is the true mean weight of all coffee jars. 2. Plan: 1-sample t-test for mean. Random sample stated; $16 < 10\%$ of all jars; population stated as approximately normal. 3. Do: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{197.5 - 200}{4.0/\sqrt{16}} = \frac{-2.5}{1.0} = -2.50 \quad (df = n - 1 = 15)$. From t-distribution table with $df = 15$: $\text{p} - \text{value} = P(T < -2.50) \approx 0.0123$. 4. Conclude: Because $\text{p} - \text{value} = 0.0123 < \alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean weight of coffee jars is less than 200 grams.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: What is a sampling distribution?

A population has mean $\mu = 100$ and standard deviation $\sigma = 60$. For samples of size $n = 100$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 6 ✓

b. 5

c. 7

d. 60

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{60}{\sqrt{100}} = \frac{60}{10} = 6.$

22 [Foundational] • Target: Mean and standard deviation of the sampling distribution

A population has mean $\mu = 100$ and standard deviation $\sigma = 117$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 10

b. 117

c. 9 ✓

d. 8

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{117}{\sqrt{169}} = \frac{117}{13} = 9.$

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23 [Foundational] • Target: The Central Limit Theorem

A population has mean $\mu = 100$ and standard deviation $\sigma = 52$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 4 ✓

b. 5

c. 3

d. 52

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{52}{\sqrt{169}} = \frac{52}{13} = 4.$

24 [Foundational] • Target: Applying the Central Limit Theorem

A population has mean $\mu = 100$ and standard deviation $\sigma = 21$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 2

b. 3 ✓

c. 4

d. 21

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{21}{\sqrt{49}} = \frac{21}{7} = 3.$

25 [Foundational] • Target: Sampling distribution of a sample proportion

A population has mean $\mu = 100$ and standard deviation $\sigma = 84$. For samples of size $n = 36$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 14 ✓

b. 13

c. 15

d. 84

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{84}{\sqrt{36}} = \frac{84}{6} = 14.$

26 [Foundational] • Target: Comparing sampling variability across sample sizes

A population has mean $\mu = 100$ and standard deviation $\sigma = 140$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 140

b. 20 ✓

c. 21

d. 19

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{140}{\sqrt{49}} = \frac{140}{7} = 20.$

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27 [Foundational] • Target: Applying the sampling distribution of the sample mean

A population has mean $\mu = 100$ and standard deviation $\sigma = 49$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 49

b. 8

c. 7 ✓

d. 6

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{49}{\sqrt{49}} = \frac{49}{7} = 7.$

28 [Foundational] • Target: Sample size and the CLT threshold

A population has a true proportion $p = 0.40$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0346 ✓

b. 0.0246

c. 0.2400

d. 0.0446

$\frac{0.40(0.60)}{200} = 0.0012.\ \sigma_{\hat{p}} = \sqrt{0.0012} \approx 0.0346.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Standard error vs. standard deviation

A population has a true proportion $p = 0.80$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0566 ✓

b. 0.1600

c. 0.0466

d. 0.0666

$\frac{0.80(0.20)}{50} = 0.0032.\ \sigma_{\hat{p}} = \sqrt{0.0032} \approx 0.0566.$

30 [Intermediate] • Target: The Central Limit Theorem (CLT)

A population has a true proportion $p = 0.70$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0365

b. 0.0265 ✓

c. 0.0165

d. 0.2100

$\frac{0.70(0.30)}{300} = 0.0007.\ \sigma_{\hat{p}} = \sqrt{0.0007} \approx 0.0265.$

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31 [Intermediate] • Target: Mean and Standard Error of the Sample Mean

A population has a true proportion $p = 0.10$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0324

b. 0.0524

c. 0.0424 ✓

d. 0.0900

$\frac{0.10(0.90)}{50} = 0.0018$. $\sigma_{\hat{p}} = \sqrt{0.0018} \approx 0.0424$.

32 [Intermediate] • Target: Sampling Distribution of Sample Proportion p-hat

A population has a true proportion $p = 0.30$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.2100

b. 0.0365

c. 0.0165

d. 0.0265 ✓

$\frac{0.30(0.70)}{300} = 0.0007$. $\sigma_{\hat{p}} = \sqrt{0.0007} \approx 0.0265$.

33 [Intermediate] • Target: Calculations with Sampling Distributions

A population has a true proportion $p = 0.30$. For samples of size $n = 500$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0305

b. 0.0205 ✓

c. 0.2100

d. 0.0105

$\frac{0.30(0.70)}{500} = 0.0004$. $\sigma_{\hat{p}} = \sqrt{0.0004} \approx 0.0205$.

34 [Intermediate] • Target: What is a sampling distribution?

A population has a true proportion $p = 0.20$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0566 ✓

b. 0.1600

c. 0.0666

d. 0.0466

$\frac{0.20(0.80)}{50} = 0.0032$. $\sigma_{\hat{p}} = \sqrt{0.0032} \approx 0.0566$.

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35 [Intermediate] • Target: Mean and standard deviation of the sampling distribution

A population has mean $\mu = 112$ and standard deviation $\sigma = 165$ (population shape unknown). For a random sample of $n = 121$, find the probability the sample mean is above 127, using the Central Limit Theorem.

a. 84.13%

b. 19.87%

c. 11.87%

d. 15.87% ✓

Since $n = 121 \geq 30$, the CLT justifies treating $\bar{x}$ as approximately normal. $\sigma_{\bar{x}} = 165/\sqrt{121} = 15$. $z = (127 - 112)/15 = 1$. $P(\bar{x} > 127) = P(Z > 1) \approx 15.87\%$.

36 [Intermediate] • Target: The Central Limit Theorem

A population has mean $\mu = 88$ and standard deviation $\sigma = 105$ (population shape unknown). For a random sample of $n = 49$, find the probability the sample mean is below 118, using the Central Limit Theorem.

a. 97.72% ✓

b. 2.28%

c. 94.72%

d. 93.72%

Since $n = 49 \geq 30$, the CLT justifies treating $\bar{x}$ as approximately normal. $\sigma_{\bar{x}} = 105/\sqrt{49} = 15$. $z = (118 - 88)/15 = 2$. $P(\bar{x} < 118) = P(Z < 2) \approx 97.72\%$.

37 [Intermediate] • Target: Applying the Central Limit Theorem

A pollster samples $n = 1200$ voters, from a population with true proportion $p = 0.70$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.

a. 0.7000

b. 0.0052

c. 0.0132 ✓

d. 0.0212

$np = 1200(0.70) = 840.00 \geq 10$. $n(1 - p) = 1200(0.30) = 360.00 \geq 10$. Both conditions hold. $p(1 - p)/n = 0.70(0.30)/1200 = 0.0002$. $\sigma_{\hat{p}} = \sqrt{0.0002} \approx 0.0132$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Sampling distribution of a sample proportion • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A pollster samples $n = 800$ voters, from a population with true proportion $p = 0.60$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1-p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.

a. 0.0253

b. 0.6000

c. 0.0173 ✓

d. 0.0093

$np = 800(0.60) = 480.00 \geq 10$. $n(1-p) = 800(0.40) = 320.00 \geq 10$. Both conditions hold. $p(1-p)/n = 0.60(0.40)/800 = 0.0003$. $\sigma_{\hat{p}} = \sqrt{0.0003} \approx 0.0173$.

39 [Challenging] • Target: Comparing sampling variability across sample sizes • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A population has standard deviation $\sigma = 60$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 5.

a. 144 ✓

b. 132

c. 12

d. 156

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$: $5 = \frac{60}{\sqrt{n}}$, so $\sqrt{n} = \frac{60}{5} = 12$, giving $n = 12^2 = 144$.

40 [Challenging] • Target: Applying the sampling distribution of the sample mean • Cross-Lesson: 3.03 The Normal Distribution

A population has standard deviation $\sigma = 152$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 8.

a. 342

b. 19

c. 380

d. 361 ✓

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$: $8 = \frac{152}{\sqrt{n}}$, so $\sqrt{n} = \frac{152}{8} = 19$, giving $n = 19^2 = 361$.

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41 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

What does the Central Limit Theorem state about the sampling distribution of the sample mean $\bar{x}$?

a. When sample size n is sufficiently large ($n \geq 30$), the sampling distribution of $\bar{x}$ is approximately normal, regardless of the population shape ✓

b. As n increases, the population becomes normally distributed

c. The sample standard deviation approaches the population standard deviation

d. Every random sample will have an identical mean

The CLT states that for sufficiently large n (conventionally $n \geq 30$), the sampling distribution of $\bar{x}$ approaches normality even if the underlying population is skewed or non-normal.

42 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Does the Central Limit Theorem apply to the shape of the individual sample data values?

a. No, it applies only to the distribution of sample statistics across repeated sampling, not the sample data itself ✓

b. Yes, individual observations become normal as n grows

c. Only if the data has no outliers

d. Yes, this is known as the Law of Large Numbers

The CLT describes the sampling distribution of the statistic $\bar{x}$, never the distribution of individual data points in a single sample.

43 [Challenging] • Target: Mean and Standard Error of the Sample Mean • Cross-Lesson: 1.01 Measures of Center & Spread

When using the formula $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, what condition must be satisfied if sampling without replacement?

a. The 10% condition: sample size $n \leq 0.10N$ (population size) ✓

b. The Large Counts Condition: $np \geq 10$

c. $n \geq 30$

d. The population must be symmetric

The 10% condition ($n \leq 0.10N$) ensures that sampling without replacement is close enough to independent that the standard error formula remains valid.

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44 [Challenging] • Target: Sampling Distribution of Sample Proportion p-hat • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Which conditions justify modeling the sampling distribution of $\hat{p}$ with a normal distribution?

a. $np \geq 10$ **and** $n(1-p) \geq 10$ ✓

b. $n \geq 30$

c. $n \leq 0.10N$

d. $p = 0.50$

The Large Counts Condition requires both expected successes $np \geq 10$ and expected failures $n(1-p) \geq 10$.

45 [Challenging] • Target: Calculations with Sampling Distributions • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A tire manufacturer knows tire lifespan is normal with $\mu = 50000$ miles and $\sigma = 4000$ miles. What is the probability that an SRS of 16 tires has a mean lifespan under 48000 miles?

a. $P(Z < -2.0) \approx 0.0228$ ✓

b. $P(Z < -0.5) \approx 0.3085$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.0500

$\sigma_{\bar{x}} = \frac{4000}{\sqrt{16}} = 1000$. $z = \frac{48000 - 50000}{1000} = \frac{-2000}{1000} = -2.0$. Tail area is $P(Z < -2.0) \approx 0.0228$.

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